Shiyue Li

Research interests: Algebraic Geometry, Combinatorics, Tropical Geometry.

Employment

University of Michigan, Ann Arbor, MI 2025–present
Donald J. Lewis Research Assistant Professor (Mentor: David Speyer)

Institute for Advanced Study, Princeton, NJ 2024–2025
Member (Mentor: Federico Ardila-Mantilla, June Huh)

Institute for Advanced Study, Princeton, NJ 2023–2024
Member (Mentor: June Huh, Jacob Lurie)

Education

Brown University, Providence, RI 2019–2023
Ph.D. and M.Sc. in Mathematics (Advisor: Melody Chan)
Thesis: K-rings and Hodge structures in graphs, matroids, and moduli spaces of curves

Yale University, New Haven, CT 2017 –2019
M.Sc. in Mathematics, Lang Fellow

Harvey Mudd College, Claremont, CA 2013 –2017
B.Sc. in Mathematics with Honors and Distinction (Advisor: Dagan Karp)
Thesis: Chow rings of heavy/light Hassett spaces via tropical geometry

Preprints & Publications

Chow polynomials and chain polynomials of matroids are not log-concave
(with Gergely Berczi, Basile Coron)
arXiv:2610.17135

Magic positivity of Snapper polynomials for matroids
arXiv:2609.24917

Matroid analogues of Gal's conjecture
(with Basile Coron, Luis Ferroni)
arXiv:2604.04550

Structural properties of nested set complexes
(with Basile Coron, Luis Ferroni)
arXiv:2601.05188

Chow polynomials of rank-uniform labeled posets
(with Basile Coron, Luis Ferroni)
arXiv:2511.13819

Kapranov degrees
(with Joshua Brakensiek, Christopher Eur, Matt Larson)
*International Mathematics Research Notices. 2025 (**20**)*

Multimatroids and rational curves with cyclic action
(with Emily Clader, Chiara Damiolini, Christopher Eur, Daoji Huang.)
*International Mathematics Research Notices. 2024 (**12**)*

K-rings of wonderful varieties and matroids
(with Matt Larson, Sam Payne, Nicholas Proudfoot.)
*Advances in Mathematics **441**. (2024)*

Equivariant log-concavity of independence sequences of claw-free graphs
*Séminaire Lotharingien de Combinatoire **89**B, 6. (2023)*

Equivariant log-concavity of graph matchings
*Algebraic Combinatorics **6**(3) (2023)*

Relative Bott–Samelson varieties
*La Matematica **2** (2023)*

Topology of tropical moduli spaces of weighted stable curves in higher genus
(with Siddarth Kannan, Stefano Serpente, Claudia Yun)
*Advances in Geometry, **23**(3) (2023)*

Wonderful compactifications and rational curves with cyclic action
(with Emily Clader, Chiara Damiolini, Rohini Ramadas)
*Forum of Mathematics, Sigma **11** (2022)*

Permutohedral complexes and rational curves with cyclic action
(with Emily Clader, Chiara Damiolini, Daoji Huang, Rohini Ramadas)
manuscripta mathematica (2022)

Intersecting psi-classes on tropical Hassett spaces
(with Marvin Anas Hahn)
*Combinatorial Theory **2**(3) (2022)*

Chow ring of heavy/light Hassett spaces via tropical geometry
(with Siddarth Kannan, Dagan Karp)
*Journal of Combinatorial Theory, Series A **178**C (2021)*

Fellowships, Grants & Awards

NSF Grant DMS-1926686, Institute for Advanced Study	2023–2025
Coline M. Makepeace Fellowship, Brown University	2021–2022

Robert Willets Carle Scholarship, Yale University 2018–2019

Kenneth and Mary Wang Fellowship, Yale University 2017–2018

Lang Fellowship, Department of Mathematics, Yale University 2017–2019

Harvey S. Mudd Merit Award (annually), Harvey Mudd College 2013–2017

Yu Yuen Kit So International Scholarship, Harvey Mudd College 2013–2017

Dean's List, Harvey Mudd College (all semesters) 2013–2017

China National Linguistics Olympiad 3rd Place 2013

China National Linguistics Olympiad Team 2nd Place 2013

Invited Talks

LG&TBQ Aug 2027
Bielefeld, Germany

JMM-AWM Special Session on "*WIAG: Topics in Algebraic Geometry*" Jan 2027
Chicago

Workshop on "*Enumerative Geometry of Matroids*" Dec 2026
Oberwolfach, Germany

ALGECOM Nov 2026
University of Illinois Urbana-Champaign

Combinatorics Seminar Sep 2026
University of Michigan

Workshop on "*Algebraic Combinatorics in Pisa*" Jun 2026
Università di Pisa

Workshop on "*Algebra and Geometry of Matroids*" Sep 2026
Banff International Research Station of Casa Matemática Oaxaca, Mexico

Institut des Sciences mini-course on "*Geometry and Combinatorics of Moduli Spaces*" May 2026
Université du Québec à Montréal

Algebraic Geometry Seminar Apr 2026
University of Michigan

AMS Special Session on "*Geometry of Moduli Spaces*" Mar 2026
Boston College

Algebraic Geometry Seminar Apr 2025
University of Pennsylvania

AMS Special Session on "*Cohomology of Arithmetic Groups, Mapping Class Groups, and Moduli Spaces*" Jan 2025
Joint Mathematics Meetings, Seattle, USA

Workshop on "*Geometry of Matroids*" Oct 2024
Institute for Advanced Study

"What is...?" Seminar Oct 2024
Institute for Advanced Study

Workshop on "*Moduli spaces of curves, moduli spaces of graphs and graph complexes*" Sep 2024
ETH Zürich, Switzerland

Workshop on "*Matroid Theory*" Jul 2024
Suzhou University, Suzhou, China

Workshop on "*Arrangements, Matroids and Logarithmic Vector Fields*", Jun 2024
Oberwolfach, Germany

Algebraic Geometry Seminar Apr 2024
Columbia University

Algebra Seminar Apr 2024
Tufts University

"What is...?" Seminar Mar 2024
Institute for Advanced Study

Mathematical Conversations Mar 2024
Institute for Advanced Study

Combinatorics Seminar (online) Feb 2024
University of Minnesota

F_1 *World Seminar (online)* Jan 2024

Special Session on "*Combinatorial perspectives on algebraic curves and their moduli*" Jan 2024
Joint Mathematics Meetings, San Francisco, USA

Preparatory Lectures for *Chow Lectures* Oct 2023
Max Planck Institute, Leipzig, Germany

Workshop on "*Curves: Algebraic, Tropical, and Logarithmic*" Aug 2023
Banff International Research Station, Canada

The 35th Formal Power Series and Algebraic Combinatorics (FPSAC 2023) Jul 2023
University of California, Davis

Workshop on "*GEMS in Combinatorics*" Mar 2023
American Institute of Mathematics, USA

Workshop on "*Algebraic Aspects of Matroids*" Banff, Canada Mar 2023
Banff International Research Station, Canada

Lecture series on "*Growing up in Sciences*" Feb 2023
Bowdoin College

Association of Women in Mathematics Student Seminar (online) Feb 2023
University of Pittsburgh

Combinatorics Seminar Feb 2023
University of Michigan

Matroid Seminar (online) Dec 2022
Fields Institute

Algebra and Algebraic Geometry Seminar Nov 2022
University of Washington

MIT-Harvard-MSR Combinatorics Seminar Nov 2022
Harvard University

AMS Special Session on "*Combinatorial Algebraic Geometry*" Oct 2022
University of Massachusetts, Amherst

AWM Research Symposium Jun 2022
University of Minnesota

RATCOW Seminar Jun 2022
University of Oregon

BATMOBILE Jun 2022
Brown University

Combinatorial and Nonlinear Day Apr 2022
Brown University

Joint Mathematics Meetings AWM Special Session (online) Apr 2022

Canadian Math Society: Combinatorics and Geometry of Moduli Spaces (online) Dec 2021

Front Range Algebraic Geometry and Number Theory Sep 2021
Colorado State University

SIAM Conference on Applied Algebraic Geometry (online) Aug 2021

Moduli Across the Pandemic (online) Apr 2021

Tropical Geometry in Frankfurt (TGiF) Seminar (online) Apr 2021

AMS Spring Eastern Sectional Meeting on "*Moduli of Curves*" (online) Mar 2021

Algebra Seminar (online) Nov 2020
Brown University

American Graduate Student Algebraic Geometry Seminar (online) Oct 2020

Service

Referee for

Advances in Mathematics
Algebraic Combinatorics (×2)
Duke Mathematical Journal
European Journal of Combinatorics
Forum of Mathematics, Sigma (×2)
International Mathematics Research Notices (×4)
Journal of the London Mathematical Society (×3)
Journal für die reine und angewandte Mathematik (Crelle's Journal)*
Mathematische Zeitschrift†
Proceedings of the American Mathematical Society (×2)
Proceedings of the London Mathematical Society
Séminaire Lotharingien de Combinatoire (Proceedings of FPSAC Conferences)† (×3)

(* means quick opinions only; † means quick opinion and/or detailed report)

Organization

University of Michigan AWM Advisory Board 2025-2026
University of Michigan Combinatorics Lunch, co-organizer 2025-
IAS Graph Complexes Learning Seminar, Organizer Fall 2024
IAS School of Mathematics Monthly Women's Lunch, co-organizer 2023–2025
IAS "What is...?" Seminar, Co-organizer 2023–2025
AMS Student Chapter at Brown University, Chair 2019–2021
Diversity and Inclusiveness Committee at Brown University, Member 2019–2020
Directed Reading Program at Brown University, Organizer 2019–2021
Directed Reading Program at Yale University, Co-organizer with Fernando Al Assal 2017–2018

Intersection: Womxn in Mathematics at Yale University, Co-organizer 2018
Yale Graduate and Professional Student Senate, Senator 2017–2018
Harvey Mudd College Math Club, President 2016–2017

Teaching

University of Michigan, Instructor
Math 565 Combinatorics and Graph Theory (Graduate) Fall 2026
Math 566 Combinatorial Theory (Graduate) Winter 2026
Math 275 Introduction to Cryptography (Inquiry-Based Learning) Fall 2025

Brown University Sheridan Teaching Certificate I Fall 2021

Brown University, Teaching Assistant
Math 90 Introduction to Calculus Spring 2023
Math 100 Calculus Part II Fall 2022
Math 100 Calculus Part II Spring 2022
Math 100 Calculus Part II Spring 2021
Math 100 Calculus Part II Fall 2020

Canada/USA Mathcamp, Mentor Summer 2019
Introduction to Group Theory
Young tableaux and Representation Theory
Young tableaux and Combinatorics
Young tabelaux and Probability

Canada/USA Mathcamp, Mentor Summer 2018
Algebraic Number Theory
Geometric Group Theory
Introduction to Modular Forms
Tropical Plane Curves
Rational Points on Elliptic Curves

Yale University, Teaching Assistant
Math 225 Linear Algebra and Matrix Theory Fall 2018
Math 112 Single Variable Calculus Spring 2018

Mentoring

LoG(M) at the University of Michigan Fall 2026
Edward (Weizhen) Tong, Co-mentor: Bailee Zacovic.
Chow functions of unbounded posets.

Reading course at the University of Michigan Summer 2026
Yujie Guo, Kate Krotkov. Co-mentor: Sam Payne
Introduction to toric varieties

Summer Undergraduate Math Research at Yale 2020
Dawit Mengesha, Robert Miranda, Brian Sun. Co-mentor: Max Kutler.
Topological zeta functions of matroids.

Canada/USA Mathcamp Learning Projects Summer 2019
Ilaria Seidel.

Alice Jenkins, Jeremy Zhou. *Schubert calculus.*
Reed Jacobs. *Primes of the form $x^2 + ny^2$.*
Simran Khunger, Rupert Li. *Enumerative geometry and Young tableaux*
Serena An. *Group actions and Sylow's theorems.*

Brown Mathematics Directed Reading Program 2019
Deven Carmichael, Joseph Hlavinka. *Ideals, Varieties and Algorithms.*
Dominick Joo, Elliott Lehrer, Joshua Lebo. *Foundations of Algebraic Geometry.*

Women in Sciences at Yale 2018
Alara Değirmenci.

Outreach

Princeton Noetherian Rings Seminar	Apr 2024
Brown Math Slam	Apr 2023
Growing up in Sciences Lecture Series, Bowdoin College	Feb 2023
Summer Undergraduate Math Research at Yale (SUMRY) Colloquium	Jul 2020
Yale Undergraduate Mathematics Society (YUMS) Colloquium	Nov 2019
Department Undergraduate Group (DUG) Colloquium at Brown	Oct 2019

Other Scientific Positions

Google Research. Mountain View, CA. Summer 2016
Research Intern. Mentors: Caroline Pantofaru, Malcolm Slaney

Howard Hughes Medical Institute. Pomona, CA. Summer 2014
Research Assistant. Mentor: Blerta Shtylla.

Summer Science Program. Socorro, NM. Summer 2012
Asteroid 1998QS52 Orbit Determination (Harvard–Smithonian Center archived).